A

Case Report

on

PROCESS PLANNING FOR A MANUFACTURED COMPONENT

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1. Introduction

1.1 Introduction

Process planning is the systematic activity of converting a component drawing into a sequence of practical manufacturing operations. It defines the raw material, machines, cutting tools, fixtures, inspection points and operating order required to produce a component that satisfies its functional dimensions. For a manufactured component, the plan must achieve the required quality while avoiding unnecessary set-ups, handling and rework.

This case study applies process planning to a machined flanged hub used in a shaft coupling. The component has turning, boring, keyway milling and bolt-circle drilling features. The study uses representative dimensions and a batch-production assumption to demonstrate how a clear route sheet can be prepared for workshop manufacture.

1.2 Process Planning Concept

A sound plan follows the principle of creating functional datums early and using them consistently. General material removal is performed before finishing. The bore and flange face are the principal functional datums of the selected hub, so they are completed in the turning set-up before the keyway and mounting holes are machined. This controls concentricity and reduces error transfer between operations.

2. Background

2.1 Problem Statement

The hub must fit a 30 mm shaft, transmit torque through an 8 mm key and align with a mating flange through six M8 mounting holes. If the bore is not concentric with the outside diameter, the rotating assembly may run out. If the keyway or bolt circle is misplaced, assembly becomes difficult and the component may fail to perform its intended function. The engineering problem is therefore to establish an economical and controlled route from steel bar stock to a finished, inspected hub.

2.2 Component Description and Case Assumptions

The selected product is a machined flanged hub made from normalized EN8/C45 steel. The study assumes an annual demand of 12,000 parts, produced in batches of 250. A CNC lathe and a vertical machining centre are available. The dimensions shown below are illustrative planning inputs and the approved engineering drawing remains the authority for manufacture.

Parameter	Assumed value	Purpose
Product	Machined flanged hub	Illustrative manufactured component
Starting stock	90 mm diameter bar x 95 mm	Allows facing and turning allowance
Finished bore	30 H7 through	Functional shaft fit
Keyway	8 mm x 3.3 mm deep	Torque transmission
Bolt circle	6 x M8 on 68 mm PCD	Coupling attachment
Batch size	250 pieces	Practical batch-production scenario

3. Objectives

3.1 Objectives and Success Criteria

The main objectives of the case study are:

• To select a logical manufacturing sequence for the flanged hub.

• To establish Datum A as the finished flange face and Datum B as the finished bore axis.

• To reduce machining set-ups by completing all rotational features in one lathe set-up.

• To specify suitable machines, tools, fixtures and inspection methods.

• To control the bore, keyway and bolt-circle features that affect assembly.

• To prepare a route that can be used for batch production and quality records.

The success criteria are that the bore achieves 30 H7, bore-to-outside-diameter runout is not greater than 0.03 mm, all sharp edges are removed and every batch has first-off and final inspection records.

4. Methodology

4.1 Process Planning Method

The planning approach starts by reviewing the drawing and identifying material, tolerances, surface finish and critical datums. The bar is cut oversize, then rough and finish machining is carried out on a CNC lathe. The finished bore locates the workpiece on an expanding mandrel in the vertical machining centre. The keyway, bolt circle and threads are then completed before deburring, washing and final inspection.

4.2 Data Collection and Assumptions

For production release, the planner would collect the approved drawing, material certificate, machine capability information, tool catalogues, gauge availability and expected batch quantity. In this academic case, representative machining data are used to show the process plan. The route is reviewed during first-off inspection and revised if dimensions or cycle time do not meet the requirement.

Process flow: Incoming material inspection -> Saw cutting -> CNC turning and boring -> Mandrel locating -> Keyway milling -> PCD drilling and tapping -> Deburring -> Final inspection -> Packing.

5. IMPLEMENTATION

5.1 Illustrative Manufactured Component

The flanged hub is made from a 90 mm diameter steel bar. The first machining set-up faces the flange side, rough turns the external diameters, drills the pilot hole and finishes the 30 H7 bore. Producing these rotating features in one chucking protects concentricity. The part is then mounted on an expanding mandrel through the finished bore, allowing the keyway and six-hole bolt circle to be positioned from the functional axis.

5.2 Detailed Process Plan

Operation	Machine / tool	Main instruction
10 Receive stock	Material inspection	Check certificate, diameter and surface condition.
20 Saw cut	Automatic band saw	Cut blank to 95 mm including allowance.
30 Rough turn	CNC lathe	Face, rough turn ODs and pilot drill bore.
40 Finish turn and bore	CNC lathe, boring bar, reamer	Finish flange face, ODs, length and 30 H7 bore.
50 Mill keyway	VMC, 8 mm end mill	Locate on expanding mandrel; check width and depth.
60 Drill and tap PCD	VMC, drill and M8 tap	Spot drill, drill 6.8 mm and tap six holes.
70 Deburr and wash	Deburring station	Remove burrs from bore, keyway and threads.
80 Final inspection	Gauges / CMM	Verify runout, bore, keyway, PCD and appearance.

6. Conclusion

6.1 Conclusion

Process planning provides a clear link between component design and workshop manufacture. In this case, the flanged hub is produced through a controlled sequence of cutting, turning, boring, milling, drilling, tapping and inspection. Establishing the flange face and bore as functional datums early in the route reduces concentricity error and supports reliable secondary machining. The plan also defines inspection points so that nonconforming parts can be detected before packing.

6.2 Future Scope

• Replace representative cycle times with time-study data from actual production.

• Use in-machine probing and statistical process control for the bore and runout dimensions.

• Introduce a dedicated hydraulic or pneumatic fixture for higher production volumes.

• Use bar-fed turning and automated part transfer to reduce handling time.

• Develop a digital route sheet linked to tool-life and inspection records.